Paper 76 — Schedule-Free SGD Is Also Effective for Nonconvex Optimization

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General framework for online-to-nonconvex conversion

Ref 106 | S / D; R / B | Must read | Original publication

The paper in one minute

Question: Does the schedule-free SGD method have a rigorous guarantee beyond convex optimization?
Model/Method: Transform online learning regret guarantees into a nonsmooth, nonconvex stationarity guarantee; recognize a particular transformation as schedule-free SGD.
Finding: With carefully selected and partly randomized interpolation parameters, schedule-free SGD reaches a (λ, ε)-stationary point at the optimal oracle-complexity order $O(\sqrt{\lambda}\,\varepsilon^{-7/2})$ (suppressing other problem constants).
Why we care: This fills a theoretical gap left by schedule-free optimization's earlier convex analysis, but does *not* establish saddle escape or global optimality.

1. First, which stationarity notion is being guaranteed?

In smooth problems we often ask for $\|\nabla F(x)\| \le \varepsilon$. For a nonsmooth function, a classical gradient may be unavailable or uninformative. The paper studies a more flexible *distributional* notion: compare gradients at nearby points whose average location is x, while penalizing how widely those points spread.

$$\|\nabla F(x)\|^{[\lambda]} = \inf_{\mathbb{E}_p[w]=x} \left\{ \left\| \mathbb{E}_p[\nabla F(w)] \right\| + \lambda \mathbb{E}_p \|w - x\|^2 \right\}.$$

A point is (λ, ε)-stationary when the expression above is at most ε (Definition 1). *Do not replace this definition by ordinary small-gradient stationarity without extra smoothness conditions.*

2. The three schedules hidden inside schedule-free SGD

The algorithm maintains three parameter sequences, not three separate models:

$$\begin{aligned} x_t &= (1 - c_t) x_{t-1} + c_t z_t && \text{(average)}, \\ y_t &= (1 - \kappa_t) z_t + \kappa_t x_t && \text{(gradient point)}, \\ g_t &\approx \nabla F(y_t), \qquad z_{t+1} = z_t - \gamma g_t && \text{(base step)}. \end{aligned}$$

The paper's novelty is *not* inventing these sequences (Defazio et al. did so), but proving that a particular choice can inherit optimal nonconvex guarantees from an online-learning method.

3. Worked example: why x, y, and z differ

Use the pedagogical one-dimensional loss $F(w) = \frac{1}{2}(w-2)^2$; let $x_0 = z_1 = 0$, $c_t = 1/2$, $\kappa_t = 3/4$, and $\gamma = 1/2$. First,

$$x_1 = 0, \quad y_1 = 0, \quad g_1 = -2, \quad z_2 = 1.$$

Then $x_2 = \frac{1}{2}(0) + \frac{1}{2}(1) = 0.5$ and

$$y_2 = \tfrac{1}{4}(1) + \tfrac{3}{4}(0.5) = 0.625, \quad g_2 = -1.375, \quad z_3 = 1.6875.$$

The gradient is evaluated at 0.625, neither at the fast iterate 1 nor the average 0.5. This toy calculation teaches the update, not the paper's nonsmooth theorem.

4. How an online-learning guarantee becomes nonconvex progress

Online optimization controls *regret*, roughly $\sum_t \langle g_t, \delta_t - u \rangle$ compared with a comparator u. Nonconvex optimization instead needs a statement about small local gradients. The authors bridge these goals through a randomized interpolation: evaluate gradients on points between two evolving trajectories and relate the regret bound to the average change in F.

Why the interpolation matters

For a differentiable objective, the fundamental theorem of calculus gives

$$F(b) - F(a) = \int_0^1 \langle \nabla F(a + s(b-a)), b - a \rangle \, ds.$$

Sampling s uniformly makes the *expected* gradient along the line segment exactly relevant to the difference $F(b) - F(a)$. That identity explains the bridge to a nonconvex descent argument even without global convexity.

Conceptual proof path
Regret control for an auxiliary online learner ⟶ **expected objective-change control** for randomized interpolation ⟶ **distributional stationarity** of an output point. This is not a Hessian or negative-curvature argument.

5. What the main guarantee really says

Assumption 1 imposes lower boundedness, a suitable line-integral identity, and a globally bounded gradient $\|\nabla F\| \le G$. Assumption 2 requires independent unbiased stochastic gradients with variance at most σ^2. The authors establish an optimal-order complexity of

$$\boxed{T = O\big(\sqrt{\lambda}\, \varepsilon^{-7/2}\big)}$$

for their stationarity measure, with additional dependence on problem constants. The result is not an ε-second-order stationary-point guarantee.

Why this is genuinely a schedule-free SGD result

Proposition 2 reconstructs a base trajectory satisfying

$$z_{t+1} - z_t = -\frac{\eta}{1-\zeta} g_t.$$

This exposes a particular online-to-nonconvex conversion as the same three-sequence update used by schedule-free SGD. But the proof needs parameter choices *specific* to the nonconvex regime, including randomized interpolation near the running average; an arbitrary practical Schedule-Free AdamW configuration is not covered.

6. Contributions and limitations

Contributions. (i) A general conversion from online learning to nonsmooth nonconvex optimization; (ii) recovery of established conversion recipes plus new ones; (iii) an equivalence between one conversion and schedule-free SGD; (iv) an optimal-order stationarity bound under an explicit oracle model. **Limitations.**

- The guarantee targets a distributional first-order measure $\|\nabla F\|^{[\lambda]}$, not global minimization, generalization, or escaping strict saddles.
- The analysis requires globally bounded gradients and an unbiased, bounded-variance stochastic-gradient oracle, and uses parameter choices that may differ from routine implementations.
- No universal convergence theorem is established for Schedule-Free AdamW or arbitrary learning-rate schedules in neural-network training.

7. Survey connections

Defazio et al. (2024): introduced the three-sequence practical algorithm and analyzed convex optimization; the present work supplies a nonsmooth nonconvex justification. **Daneshmand/Jin:** their targets concern negative curvature or approximate second-order stationarity, whereas this work proves a different, first-order-type criterion. **Lookahead:** both use multiple weight sequences but have different mathematical goals.

8. Quick reference

Stationarity	$\|\nabla F(x)\|^{[\lambda]} \le \varepsilon$ (Definition 1).
Base iterate	$z_{t+1} = z_t - \gamma g_t$.
Average iterate	$x_t = (1 - c_t)x_{t-1} + c_t z_t$.
Gradient iterate	$y_t = (1 - \kappa_t)z_t + \kappa_t x_t$.
Assumptions	$\|\nabla F\| \le G$; unbiased oracle, variance $\le \sigma^2$.
Complexity	Optimal-order $O(\sqrt{\lambda}\,\varepsilon^{-7/2})$, plus problem constants.

Papers Master note

Ahn et al. establish a general online-to-nonconvex conversion and identify a special case as schedule-free SGD, with optimal-order convergence to (λ, ε)-stationarity under bounded-gradient/oracle assumptions. This is not a saddle-escape or global-minimum guarantee.

Original: https://proceedings.mlr.press/v267/ahn25a.html. The pedagogical scalar example is not from the paper.